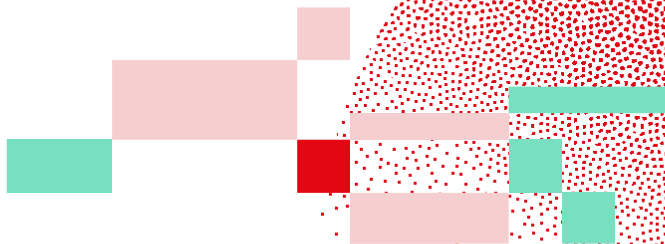




Swiss Institute of
Bioinformatics



Introduction to clustering

SIB training course

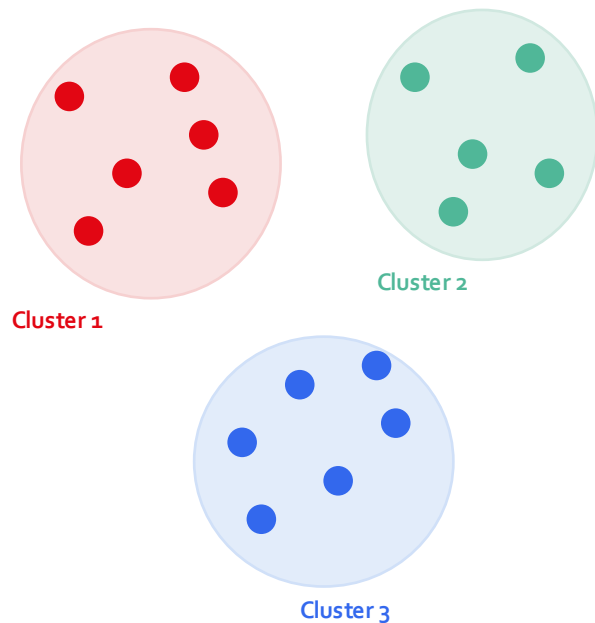
08/06/2026

What is clustering?

Clustering is an **unsupervised** task: we ask the algorithm to discover natural groupings in data — **without any labels** or prior knowledge of group identity.

Contrast with supervised learning

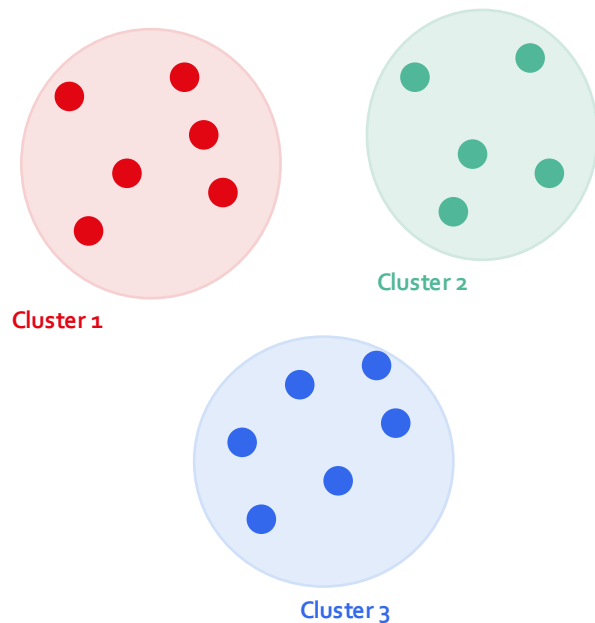
	Supervised	Unsupervised
Labels	Required	None
Goal	Predict class	Discover groups
Output	Class per sample	Cluster per sample



No labels were given — structure emerged from the data.

Why is clustering used?

Pattern detection
Compression
Dimensionality reduction
Data augmentation
Outlier/anomaly detection
Semi-supervised learning
Preprocessing for other analyses
Visualization and interpretation



No labels were given — structure emerged from the data.

How does clustering matter in biology?



Transcriptomics

Identify cell types or states in single-cell RNA-seq data without prior markers.



Microbiology

Group microbial sequences or OTUs to reveal community structure.



Patient stratification

Discover patient subgroups with distinct disease profiles or treatment responses.



Image analysis

Segment tissue regions or organelle populations in microscopy data.

From visualization to structure

PCA

linear

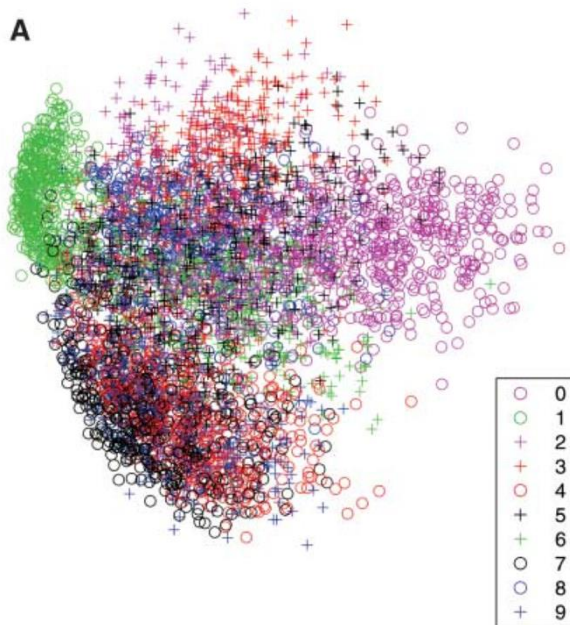
Projects data onto 2D space.
Fast, interpretable structure.



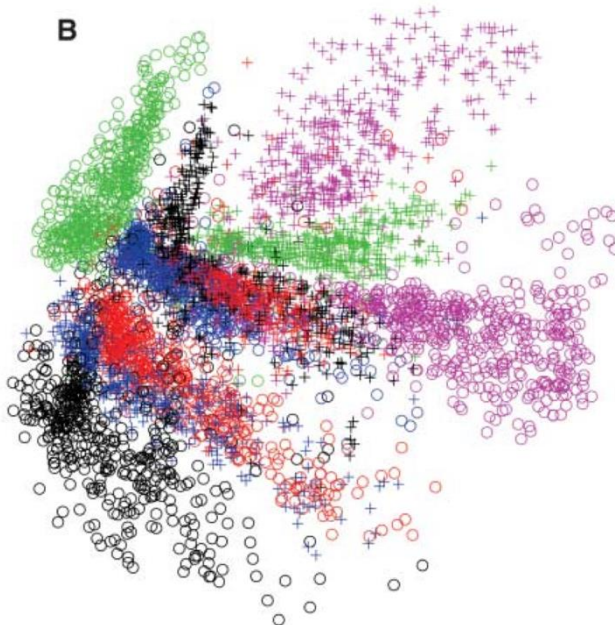
Linear projection — 1

t-SNE

non-linear



B



UMAP

non-linear

al structure. Faster
d in single-cell biology.



ster, scalable

The gap: seeing vs. assigning

Visualization (PCA, UMAP, t-SNE)

- Shows cluster-like structure in 2D
- Subjective: relies on human judgment
- Sensitive to parameter choices
- Doesn't scale to more samples
- Can introduce projection artefacts



we need
a method

Clustering algorithm

- Assigns every point to a group
- Objective, reproducible
- Quantifiable with validation metrics
- Scalable to large datasets
- Makes assumptions explicit

Key question: what does "same group" mean?

Clustering algorithms

k-Means

Hierarchical

DBSCAN

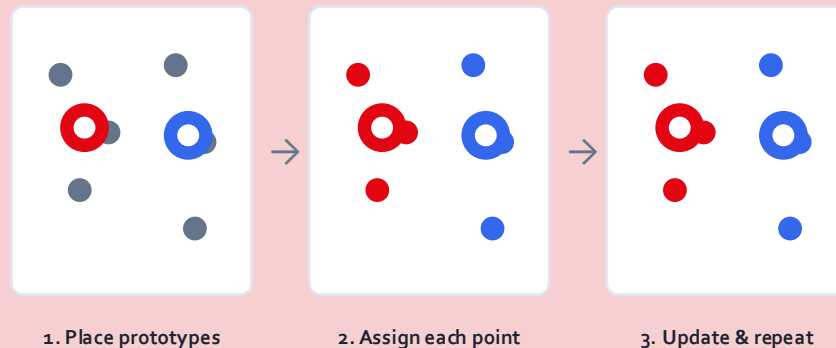
Gaussian Mixture

The intuition

Think of sorting cells by similarity to a prototype

Choose k representative "prototype" cells and assign every other cell to whichever prototype it resembles most.

Update each prototype to be the average of its assigned cells and repeat until the groups stabilise.



Key idea:

Each point belongs to exactly one group (hard assignment). Similarity = distance to the nearest centroid.

Biological parallel →

Grouping single cells by their nearest average expression profile.

k-Means

Centroid-based · Hard assignment

1

Choose k

Specify the number of clusters k in advance.

2

Initialise centroids

Place k centroids randomly (or with k-means++ for better starts).

3

Assign points

Each point joins the nearest centroid — Euclidean distance.

4

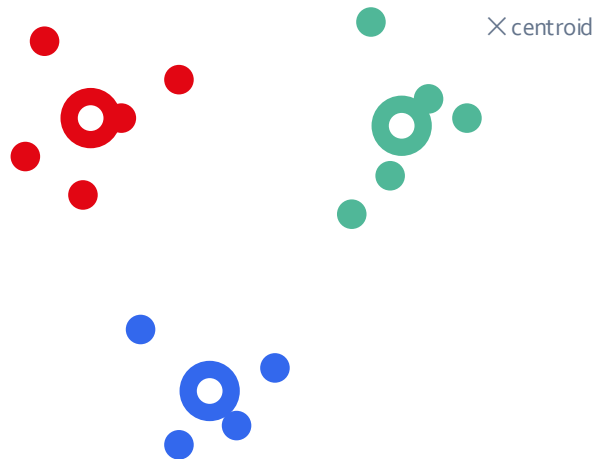
Update centroids

Move each centroid to the mean position of its cluster.

5

Repeat

Iterate steps 3–4 until assignments stop changing.



Strengths

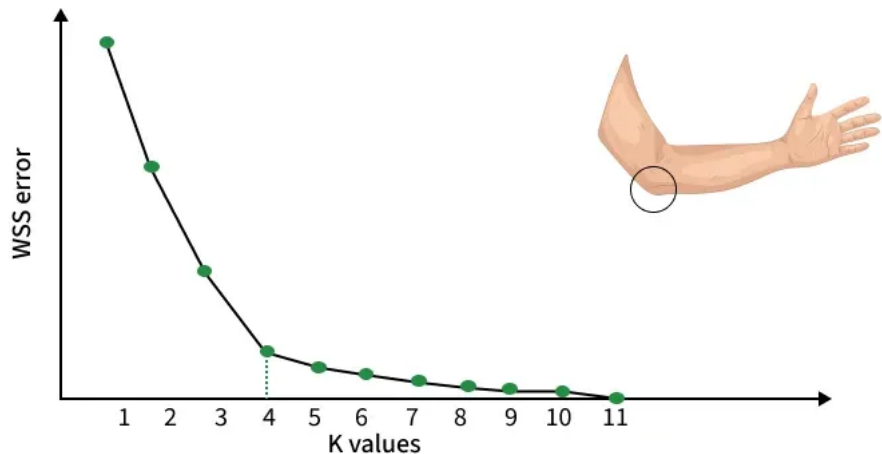
Fast and scalable
Simple to understand

Limitations

Requires k upfront
Assumes spherical clusters
Sensitive to initialisation

How to choose k?

Elbow method



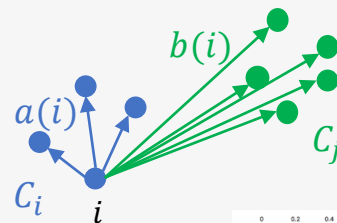
Elbow: look for the "bend"

Silhouette

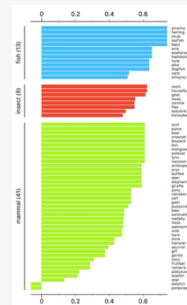
$$a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i, i \neq j} d(i, j)$$

$$b(i) = \min_{j \neq i} \frac{1}{|C_j|} \sum_{l \in C_j} d(i, l)$$

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$



Silhouette coefficient: $\max_k \langle s(i) \rangle_i$



Silhouette: higher is better

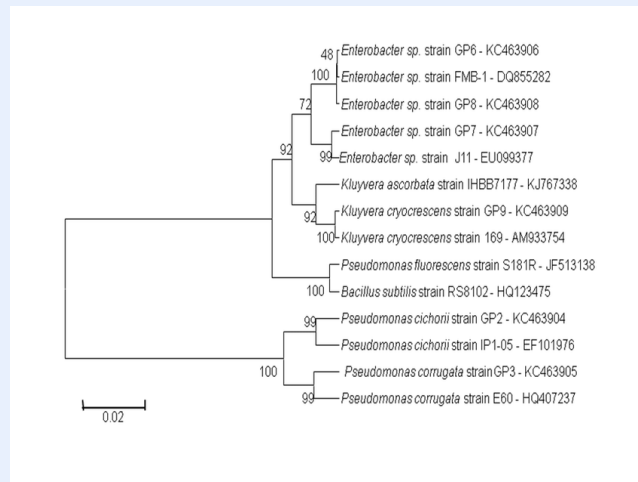
The intuition

Think of building a phylogenetic tree

Every strain starts as its own group.
 The two most similar strains merge first.
 Then the next closest pair merges, and so on up to one big group.
 The result is a tree showing every level of relationship at once.

Taxonomy:

Species → Genus → Family → Order
 merge by similarity at each level.



Key idea:

It is not decided upfront. The full tree is cut at any level to produce the number of clusters that makes biological sense.

Hierarchical Clustering

Agglomerative (bottom-up) · Dendrogram

Start: each sample is its own cluster.

Merge: at each step, join the two closest clusters.

Repeat: until all samples belong to a single cluster.

Result: a tree (dendrogram) encoding every level of grouping.

Linkage criteria

Single: Shortest distance between any two points in clusters

Average: Average distance between any two points in clusters

Complete: Longest distance — compact, equal-size clusters

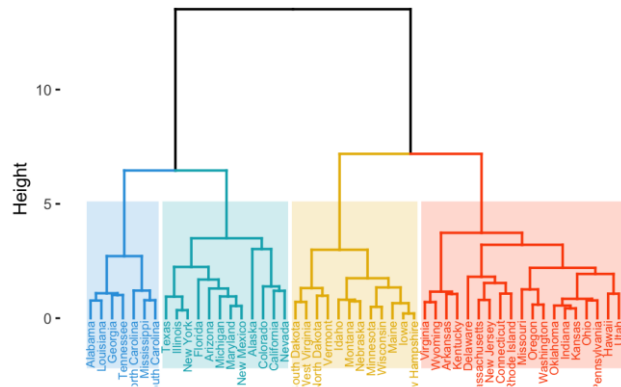
Ward: Minimises within-cluster variance — often best choice

- ✓ No need to specify k beforehand
- ✓ Dendrogram reveals nested structure
- ✗ Computationally heavy for large n ($O(n^2)$)
- ✗ Greedy — merges cannot be undone

Cluster Dendrogram

Height

Alabama Louisiana Tennessee Mississippi North Carolina South Carolina Illinois New York Florida Alabama Michigan New Mexico Alaska California Nevada Nevada Utah Virginia West Virginia North Dakota South Dakota Montana Wyoming Idaho Minnesota Wisconsin Wisconsin Iowa New Hampshire New York Working Arkansas Arkansas Delaware Delaware Connecticut Rhode Island Oregon Washington Oklahoma Indiana Kansas Kansas Pennsylvania Pennsylvania Hawaii



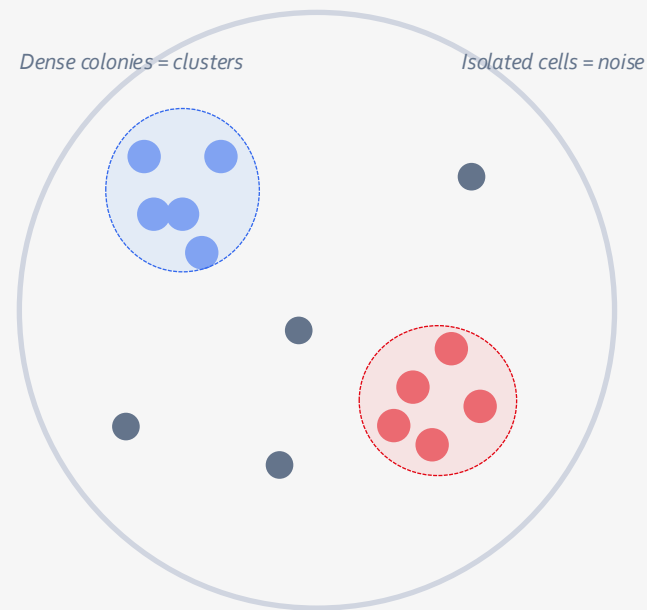
The intuition

Think of bacterial colonies on a petri dish

A cluster is wherever cells are densely packed together.

A cell belongs to a colony if it has enough neighbours nearby.

Isolated cells far from any colony are labelled noise.



Key idea:

Clusters emerge from density — any shape is allowed, and outliers are found automatically without you telling the algorithm how many groups to look for.

Biological parallel →

Detecting cell subpopulations in high-dimensional flow cytometry or spatial transcriptomics data.



DBSCAN

Density-based · No k needed · Detects outliers

Two parameters:

ϵ : neighbourhood radius

minPts: minimum points to form a dense region



Core point

Has $\geq \text{minPts}$ neighbours within radius ϵ



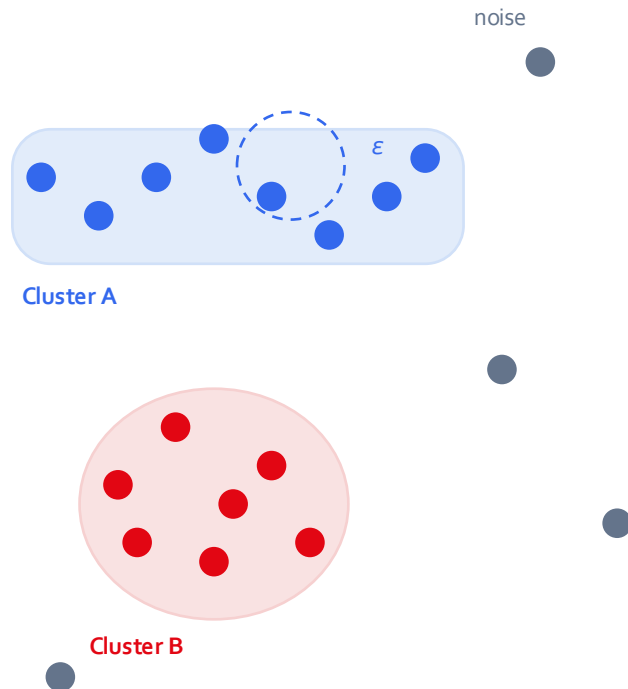
Border point

Within ϵ of a core point, but fewer than minPts neighbours



Noise / outlier

Neither core nor border — labelled as noise



- ✓ Arbitrary cluster shapes
- ✓ Built-in outlier detection
- ✗ Struggles with varying density

The intuition

Think of overlapping distributions in a mixed population

Gene expression from a sample containing two cell types mixed together.

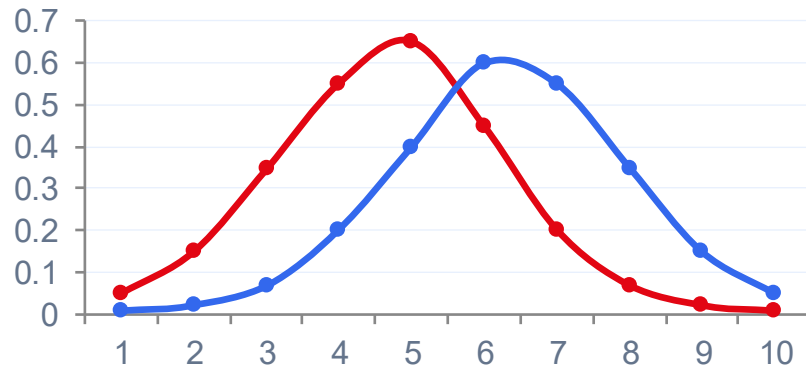
Each cell type follows its own bell-like Gaussian distribution.

GMM finds the best-fitting overlapping bells.

Unlike k-Means:

A cell in the overlap gets soft membership — e.g. 70 % Type A, 30 % Type B.

Mixed population — gene expression



Key idea:

Each cluster is a Gaussian bell.

Membership is probabilistic — cells in ambiguous regions belong to multiple clusters with some degree of uncertainty.

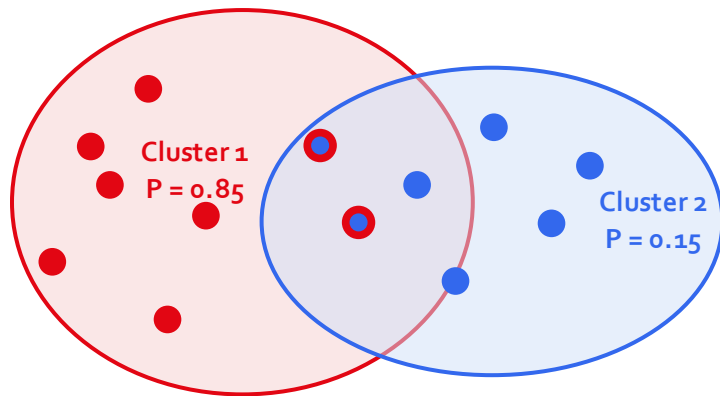
Gaussian Mixture Models

Probabilistic · Soft assignment · Fitted by EM

Idea: assume the data is generated by a mixture of k Gaussian distributions. Each component has its own **mean**, **variance**, and **mixing weight**.

Key distinction vs k-Means

	k-Means	GMM
Assignment	Hard (0 or 1)	Soft (probability)
Cluster shape	Spherical	Elliptical (flexible)
Output	Label	Probability per cluster
Uncertainty	None	Quantified



*Overlap region:
shared probability*

- ✓ Handles overlapping clusters
- ✓ Soft uncertainty
- ✗ Assumes Gaussian distribution

Fitted with Expectation-Maximisation (EM): alternate between assigning probabilities (E-step) and updating parameters (M-step).

Side-by-side comparison

Property	k-Means	Hierarchical	DBSCAN	GMM
Specify k?	Yes	No (cut dendrogram)	No	Yes
Assignment	Hard	Hard	Hard	Soft
Cluster shape	Spherical	Any	Any	Elliptical
Handles noise?	No	No	Yes (built-in)	No
Scales to large n	✓ Fast	✗ $O(n^2)$	✓ Good	✓ Moderate
Probabilistic?	No	No	No	Yes

No single best method — choose based on your data geometry, size, and scientific question.

How to choose an algorithm?



Do you expect compact, round clusters?

→ k-means is likely sufficient.



Do you want to explore nested structure?

→ Hierarchical with the dendrogram.



Do you have noise or outliers?

→ DBSCAN handles them natively.



Do clusters overlap? Need probabilities?

→ GMM gives soft, probabilistic assignments.



Large dataset (> 10 000 cells)?

→ k-means or DBSCAN; avoid hierarchical.



Validating clustering results

Internal metrics

(no ground truth needed)

Silhouette score

How similar is a point to its own cluster vs. others? (-1 to $+1$, higher is better)

Davies–Bouldin index (DBI)

Average ratio of within-cluster scatter to between-cluster distance. Lower is better.

Calinski–Harabasz index (CHI)

Ratio of between-cluster to within-cluster dispersion. Higher is better.

External metrics

(requires known labels — e.g. cell type annotations)

Adjusted Rand Index (ARI)

Measures agreement between predicted clusters and true labels, corrected for chance. Range: -1 to $+1$.

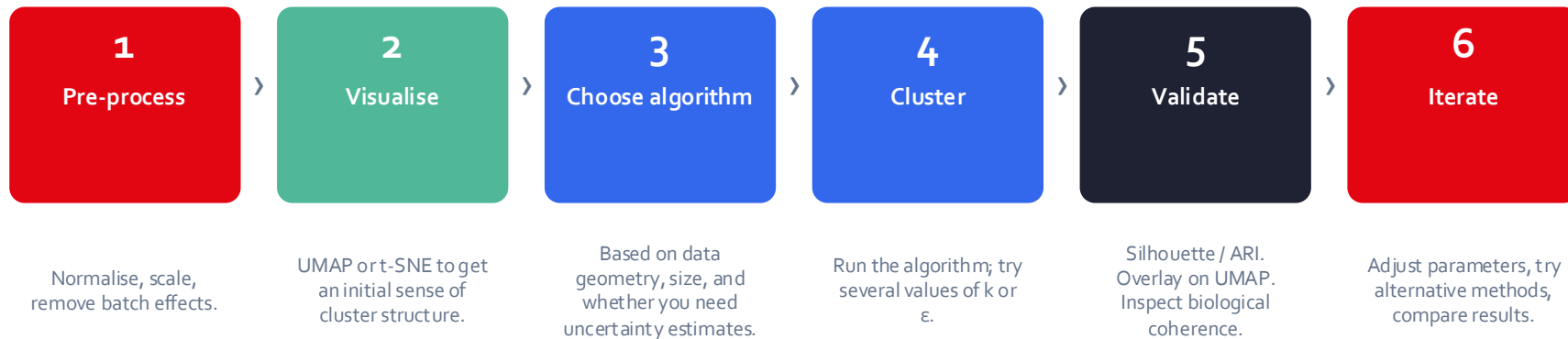
Normalised Mutual Information (NMI)

Mutual information between cluster assignments and labels, normalised to $[0, 1]$.

Tip: Always complement numerical metrics with visual inspection (UMAP / t-SNE) and biological interpretation.



Suggested pipeline



Take-home message

No universally best clustering algorithm, each method embeds a different notion of similarity.

Match the algorithm to your data geometry, validate results numerically and biologically.

